

Semester III (B.Tech.)

No. of Printed Page 1

Er. No.

Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna

T-1 (Odd Semester 2022)

18B19GE399-Environmental Science

Max. Duration: 1 Hour

Max. Marks: 15

Note: 1. This question paper has 03 questions.

2. Write relevant answer only.

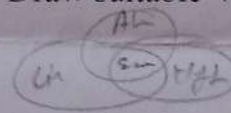
3. Do not write anything on question paper except enrolment number.

Q1. (a) Elaborate the consumptive and non-consumptive value of "Forest Resource" [3]

(b) Define "Solar Heat Collectors" and list out its major advantages. [2]

Q2 What do you mean by "Ecological Succession"? Elucidate various steps of the process of the ecological succession. [5]

Q3. (a) What do you mean by "Sustainable Development"? Draw suitable Venn diagram of sustainable development and explain it. [3]



(b) Elucidate the thermal stratification of atmosphere and also show the pattern of temperature change. [2]



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Semester III (B.Tech)

Er. No.....
Academic Year: 2022-23**Jaypee University of Engineering & Technology, Guna****T-1 (Odd Semester 2022)****18B11HS312 - TECHNIQUES FOR DECISION MAKING**

Maximum duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 04 questions.
2. Write relevant answers only.
3. Do not write anything on question paper.
4. Use of calculators is permitted.

Q1. Describe the stages of decision making process with the help of an appropriate diagram. [05]

Q2. A charitable organization decided to give old age pensions to people over sixty years of age. [05]
The scale of pensions were fixed as shown below:

Age Group (Years)	Pension Amount
60—64	Rs. 200 per month
65—69	Rs. 250 per month
70—74	Rs. 300 per month
75—79	Rs. 350 per month
80—84	Rs. 400 per month

The ages of 25 persons who secured the pension right are as given below:

74, 62, 84, 72, 61, 83, 72, 81, 64, 71, 63, 61, 60, 67, 74, 64, 79, 73, 75, 76, 69, 68, 78, 66, 67

Calculate the monthly average pension payable per person. Find the variance of the distribution.

Q3. If the median of the distribution given below is 28.5, then compute the value of x and y. [03]

Group	1-10	11-20	21-30	31-40	41-50	51-60	Total
Frequency	5	x	20	15	y	5	60

Q4. Find the standard deviation of the following data set: [02]

32, 6, 21, 13, 8

Jaypee University of Engineering & Technology, Guna

T-1 (Odd Semester 2022)

18B11EC311 – DIGITAL SYSTEMS AND MICROPROCESSORS

Maximum Duration: 1 Hour

Maximum Marks: 150

Notes:

1. This question paper has five questions.
2. Write relevant answers only in proper order.
3. Do not write anything on question paper (Except your Er. No.).
4. Calculator is not allowed.

61500
12 40
2400
480

- | | Mark |
|---|-------|
| Q1. (a) Determine the base of the numbers for the following operation to be correct,
$(34)_r + (12)_r = (30)_r$ | [01] |
| (b) Convert $(586739.625)_{10}$ to its equivalent binary number. | [01] |
| (c) Convert $(232347561435)_8$ to its equivalent hexadecimal number. | [01] |
| Q2. (a) Find Subtraction $(402)_8 - (314)_8$ using 7's and 8's complement method.
001000110010001101000111 | [1.5] |
| (b) Find the subtraction $(9)_{10} - (5)_{10}$ in 2's complement form. | [1.5] |
| Q3. (a) Convert $(7038)_{10}$ into excess-3 code. 00001111 | [1.5] |
| (b) A seven-bit Hamming code is received as 1001010 at the receiver. Find whether the transmitted code is correct or not. If there is an error, find out the correct code transmitted and the data bits from the corrected code. | [1.5] |
| Q4. For the following Boolean function: $F = A'BC + AB'C' + A'B'C' + AB'C + ABC$ | [03] |
| (i) Find the truth table | |
| (ii) Use Boolean algebra to simplify the function to minimum number of literals. | |
| (iii) Draw the logic diagram using basic gates. <i>minimized expression</i> | |
| Q5. Find the minimized expression using K-Map,
$F = \overset{1}{AB'C'D} + \overset{2}{A'BCD} + \overset{3}{A'B'C'D} + \overset{4}{ABC'D} + \overset{5}{ABCD} + \overset{6}{AB'C'D'} + \overset{7}{AB'CD'} + \overset{8}{A'BC'D} + \overset{9}{A'B'C'D'} + \overset{10}{A'B'CD'}$ | [03] |

Semester III (B.Tech.)

Er. No.....
Academic Year: 2022-23**Jaypee University of Engineering & Technology, Guna****T-1 (Odd Semester 2022)****18B11CI311 – DATA STRUCTURES**

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 03 questions.
2. Write relevant answers only.
3. Do not write anything on the question paper (Except your Er. No.).

Q1. Write a time-efficient searching algorithm to find an element in a sorted array of integers. Also, analyze the time complexity of your algorithm in best and worst cases. [05]

Q2. Write Selection sort algorithm and apply on the following array to arrange the elements in ascending order. Show all comparisons in each pass. [05]

6	4	14	16	13	10	11	9	4	5
4	4	5	6	9	10	4	13	16	17

Q3. Write short notes on the following: [05]

- (a) Non-linear data structures $T_n \leftarrow$
- (b) Time complexity and Asymptotic notations
- (c) Row-major order Vs. Column-major order

Jaypee University of Engineering & Technology, Guna**T-1 (Odd Semester 2022)****18B11CI312- DATABASE SYSTEMS**

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has **three** questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

- | | | Marks |
|------------|--|--------------|
| Q1. | In context to ER modelling, explain following terms with illustrative diagram:
a) Relationship instance
b) Recursive relationship | [05] |
| Q2. | Consider the following set of requirements for a company database and complete conceptual database design step using ER modeling to develop the desktop application:
→ Company develops products and these products are having name, product ID and description of the product.
→ Products are made using many components. Each component can be supplied by one or more suppliers having supplier ID, Supplier name and supplier address.
→ It is also required to keep information about components like; component ID, name, description, suppliers who supply them, and products in which they are used.
→ Company delivers developed products to various wholesalers. For wholesalers it is require to store name, id, address, product, and required quantity of the product. Where application users want to search wholesalers with city or state.
→ A supplier can exist without providing components, but a product cannot exist without components. A component does not have to be associated with a supplier and a product. Not all components are used in products. Draw neat and clean ER diagram using five steps as discussed in Database Systems' class lectures. | [05] |

Q3.

How the abstraction process in the database approach is different from the object-oriented and object-relational databases? Explain by the suitable example.

[05]